Communities of the Interested

Redistricting Institution Congruence with Communities of Interest Among Geocoded Public Comments

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1 Introduction

1.1 Redistricting and Democracy

Redistricting, a boundary-drawing process undertaken by every state in the United States at least every decade, has increasingly become a critical factor in determining the health of the nation's representative democracy (Grumbach 2023). The process of linking local constituencies to their representatives based on territory has profound downstream consequences for a variety of political outcomes, including district level competitiveness (Jasny et al. 2025; Kenny et al. 2023), voter turnout (Fraga 2016), and the creation of majority-minority and minority-influence districts (Barabas and Jerit 2004). Redistricting thus provides state governments with immense power to tip the scales towards or away from an electoral landscape that equally enfranchises the voters that make up their constituencies.

However, despite it's importance, redistricting is nonetheless at odds with traditional democratic pipelines of connecting public opinion to representation and then policy. One one hand, there's an unclear connection between each representative and their constituency based on the fact that the constituencies themselves are being shuffled into a new arrangement of districts. In other words, a representative, whether at the state legislature, state senate, or congressional level, need not necessarily concern themselves with backlash or retrospective voting because the very population that might next vote for them is subject to change. On the other hand, there's no consistent, structural way for the public to signal their desired 'vote' on a particular map. Statewide referenda on district plans are rare, and a statewide district map can empower some voters while disenfranchising others, making aggregate measures of approval more arbitrary than the mass approval of policies that are applied equitably across a state. While the likely overall partisan balance resulting from a boundary configuration can be an ideological signal to voters (McCarthy 2022), very few are likely to have the knowledge and expertise necessary to evaluate the entirety of a district map down to the precinct level, especially given low rates of geographic literacy among Americans (McFarlane 2024). This then leaves the task of translating a full plan into comprehensible public insights up to media ecosystems, interest groups, and the relevant office holders themselves.

True opinions on a map among the public are likely localized and related to the partisan or nonpartisan

factors of the communities Americans call home. A voter might not have a good sense of how to divide up an entire metro area into state house districts, but they can probably comment on whether their neighborhood, city, school district, and county should be kept whole, combined with other areas, or separated from other areas. Place based identities are a powerful force in the United States (Elder and Lueders 2025), implying that residents of a community have some perception of the social, cultural, economic, and geographic dynamics that separate "where they live" from everywhere else. It is under these conditions that the concept of "Communities of Interest" can play a particularly cogent role, providing a personal link between the individual voter and the clandestine practices of legislative district boundary drawing.

1.2 Communities of Interest in Legislative Redistricting

Nicholas Stephanopoulos (2012) articulates that a group of people constitute a community of interest (COI) if they (1) share a common geographic range, (2) share common socioeconomic or cultural interests, and (3) subjectively identify as members of a cohesive geographic community. Respecting communities of interest is typically one of several criteria that many boundary-drawing institutions are constitutionally obligated to respect (Webster 2013), including contiguity, compactness, population equality, and protection of minority voting populations. However, these criteria are difficult to balance, and some can even be contradictory. In fact, the "reapportionment revolution" of the 1960s, a series of Supreme Court decisions requiring that legislative redistricting strictly adhere to the criteria of population equality across districts, is credited with massively disrupting the ability of states to respect COIs despite the clear benefits of meeting both criteria simultaneously (Stephanopoulos 2012). State and federal courts are nonetheless receptive of arguments in favor of protecting COIs, often striking down maps in clear violation of the criteria. Furthermore, recent court decisions have diminished the prominence of the Voting Rights Act as a basis for striking down statewide maps that fail to appropriately represent minority populations (Chen et al. 2022), leaving communities of interest as a possible line of defense for these populations to still litigate with.

Despite their utility and relevance to the courts, measuring the extent and frequency of COIs has proven difficult, and no procedure has emerged as a clear national standard. Some scholars have used contextual and behavioral characteristics of smaller geographic units within states (Kruse, Gao, and Mayer 2026; Gimpel and

Harbridge-Yong 2020; Makse 2012; Syme 2025; Rossiter, Wong, and Delamater 2018; Swanson and Bryan 2024), clustering units into cohesive groups and comparing district maps to group boundaries. Kruse, Gao, and Mayer (2026) use Facebook friendship networks and mobile device visits between Census block groups to construct COIs, while Maske sorts communities based on issue area initiative votes in Ohio (Makse 2012). These observational approaches present defensible solutions based on well-distributed data across entire states, and their sophisticated methods have yielded actionable insights, but they nonetheless ascribe community membership at a distance from the public's participation in the redistricting process.

Other scholars, therefore, have constructed COIs based on grassroots input from the public, including the deployment of public mapping tools that allow respondents to "draw" their communities of interest from a bird's eye view (Chen et al. 2022; Chambers et al. 2022). These drawn communities of interest can then be compiled and analyzed using a variety of computational techniques. Chambers and colleagues cluster overlapping COIs based on their similarity across respondents, while Chen and colleagues introduce novel measures to evaluate how egregiously an enacted plan splits a given community. These studies provide a framework that boundary-drawing institutions can use to engage the public and better link redistricting with democratic participation. They also provide a closer estimation of the kind of communication flows between constituents and legislators that are likely to occur during the redistricting process.

Thankfully, many states, especially those now making use of an Independent Redistricting Commission (IRC), have undertaken significant efforts to collect large bodies of public testimony and assess localized public opinion, even specifically prompting respondents to discuss communities of interest in some cases (Rambo 2021; Subia 2024). Large databases of testimony raise questions regarding who has provided their opinions and whether this input has had any impact on the final maps being produced. With mechanisms of connecting redistricting to democratic participation growing more prominent, is is now more feasible to measure congruence between state legislative decision making and constituent preferences, especially regarding communities of interest, which can be mapped and understood in the same territorial space as the legislative districts being drawn.

1.3 Redistricting and Congruence

Extant literature has found congruence to be markedly low in most states (Lax and Phillips 2012), but the way this finding translates itself to redistricting and specifically communities of interest is still an emergent question, especially given the aforementioned disconnect between redistricting and accountability through voting. Any level of congruence during redistricting depends on the ability of concerned citizens to make their voices heard in the first place. Gabriella Subia's exploration of public participation during the creation of Colorado's eighth congressional district suggests that the state's IRC did in fact prioritize the representation of Hispanic voters, a competitive partisan balance, and the protection of various industries north of Denver, all of which were advocated for by both interest groups and local residents (Subia 2024). This case study aligns itself with the broader literature on public comment during rulemaking, which has generally found that institutions are responsive to those who participate (Sahn 2025; Ingrams 2023). However, another analysis by Olivia Rambo finds among a sample of comments submitted to the California Citizen's Redistricting Commission that certain interests were more likely to achieve their boundary drawing aims than others, and many comments were copies of templates distributed to the public by interest groups (Rambo 2021). This study is consistent with the idea that the most influential voices during the process of collecting public testimony are not representative of the population as a whole (Yackee and Yackee 2006), highlighting one source of disconnect between mass opinion and redistricting outcomes.

Incongruence with public opinion, furthermore, might not only be a function of an unrepresentative sample of public input, but also deliberate actions by boundary-drawing legislators. States have adopted a variety of map-drawing procedures, each giving partisan legislatures a different level of leeway over determining the final maps based on systems of oversight and stalemate-breaking (McCartan et al. 2026). McCartan and colleagues find that increased partisan leeway is positively related to the creation of more biased maps that are less responsive to statewide electoral shifts. Moreover, even if we assume that map-drawing institutions are listening to public input, it may be the case that only certain public requests are considered legitimate. Butler and Dynes (2016) find that office holders often discount the testimony of those with whom they disagree regarding other issue areas like economic policy. Redistricting may thus be incongruent with

both mass opinion, which is difficult to assess in the first place, and any broad corpus of public testimony, an already unrepresentative sample that legislators may filter based on interest group pressures and partisan bias.

1.4 Web Comment Testimony

When evaluating public input on redistricting, it's important to consider that certain types of testimony may be more democratized than others, especially as redistricting institutions professionalize and make use of digital communication platforms. Attending a public hearing requires, for many, a significant time commitment and some expenditure on traveling to the location of the meeting to speak. Many states also offer citizens the opportunity to submit long-form content, including essays, presentations, and even statewide district plans that the citizen can create with publicly-available software. These formats, along with public hearing attendance, might be the preferred submission type for partisan actors, interest groups, and highly-engaged members of the public who possess some level of expertise regarding the redistricting process. Alternatively, many states have offered a public web commenting platform allowing citizens to submit a brief questionnaire about their redistricting preferences, often in relative or complete anonymity. These platforms are available free of charge to anyone with broadband access, and their simplicity likely makes them a less efficient venue for interest groups and partisan actors. A web commenter can thus provide their input in minutes without needing expert-level knowledge of their state's political geography nor a comprehensive understanding of an existing or proposed map. Instead, they can focus on a specific community of interest that they know best, or perhaps another focused concern about the redistricting process.

It's unclear how seriously boundary-drawing institutions take these web comment platforms, or what sorts of requests they might entertain over others, and even these platforms are unlikely to gather opinions from a truly representative sample of the public. Nonetheless, nationwide web comment collection in the 2021-2022 redistricting cycle constitutes the most accessible and widespread efforts to date at measuring the preferences of citizens who hope to see their communities fairly represented when legislative boundaries are drawn. We therefore find it vital to explore the geographic, partisan, and racial makeup of web commenters where such information is attainable. We also hope to evaluate the congruence between legislative boundaries

drawn and the communities of interest that web commenters have requested be kept whole, combined with other communities, or separated from other communities.

2 Methods

3 Results

4 Discussion

5 Appendix

5.1 Appendix A: Comment Data Extraction with Ellmer

The `ellmer` package in R was used to interact with off-the-shelf LLM chat features and extract structured data from public testimony comments (Wickham et al. 2025). Barrie, Palmer, and Spirling (2025) recommend the use of open-weight Large Language Models (LLMs), which allow for more consistent replication of coding and classification tasks compared to the more fragile but advanced proprietary frontier LLMs. Therefore, we make use of the "Mistral Large 2.0" model, available on OpenRouter (https://openrouter.ai/), combined with a three-stage `ellmer` schema to gather comment metadata, location information, and location relationships for comments deemed . Mistral Large 2.0 was chosen for its strengths in named entity recognition and structured data extraction from administrative texts (https://mistral.ai/industry/public-sector/).

5.1.1 Stage One - COI Relevance Screening

Many states offered the public a general portal through which to comment on any redistricting concern. Some commenters thus used these portals to voice frustration or approval about traditional redistricting outcomes such as compactness, competitiveness, incumbency protection, and minority representation, while others weighed in about procedural matters, transparency, and institutional reform. In order to screen for comments concerning communities of interest that mention specific locations to geocode, the following prompt was paired with each comment, and output was constrained to a binary structured data type with `ellmer`:

> Binary indicator measuring the specificity of this {{state}} commenter's description of a community of interest. A value of 1 indicates that the commenter is clearly discussing particular, named locations that they want kept together or separate for a district map. A value of 0 indicates that the commenter fails to mention specific locations or that they are discussing a different redistricting concern.

Only comments meeting COI relevance criteria were including in downstream data extraction steps. Because

prompting was consistent across all comments, screening was conducted for each state using a single parallel structured chat call.

5.1.2 Stage Two - Comment Metadata Extraction and Named Entity Recognition (NER)

The second stage of the `ellmer` pipeline was applied to every comment meeting COI relevance criteria. In this stage, a highly-structured prompting architecture was built to identify named geographic locations that were part of a specific redistricting configuration requested by the commenter. While other geographic NER software has proven successful at similar tasks (Halterman 2023), we chose to use `ellmer` for its flexibility in extracting more detailed location information as well as its use of well-maintained, open-source models and its compatibility with other R packages used throughout our analysis. Using a parallel structured chat with `ellmer`, the following system level prompt was used to first introduce the Mistral model to the task at hand:

> You are an expert geocoding research assistant evaluating public comments from the 2021-2022 redistricting cycle in the United States for quantitative downstream evaluation of institutional compliance with constituent input. The goal is to extract consistently named and formatted locations and comment-level metadata. The provided comments were all written by concerned citizens from {{localContext}}.

The `localContext` variable was based on any state, county, or ZIP Code level identifiers available within comment level metadata. For example, Colorado comment data included commenter ZIP Codes, so "ZIP Code XXXXX in Colorado" was assigned to `localContext` for all comments from the corresponding area. Each set of comments sharing the same value of `localContext` were bundled and processed with a separate parallel structured chat call. If location context was only available at the state level, prompts were bundled by temporal strata such as the date of the comment submission.

Each specific location mentioned throughout the comment was defined by a prompt schema of its own, all of which were embedded in an array type response from `ellmer`. This schema consisted of an enum (factor) response describing the location's administrative level, a string response for the location name, an enum

response for any cardinal direction subareas or other codifiable subareas relevant to the mentioned location, and a binary response reflecting whether the location was mentioned purely as context for an argument or narrative rather than being included in a requested configuration. The following top-level prompt for location recognition also included several rules designed to catch edge cases and constrain location mentions to those specifically mentioned by the commenter in a redistricting request:

> Return individual geographic locations, including landmarks, neighborhoods, townships, boroughs, towns, cities, school districts, counties, legislative districts, and regions, in accordance with the following rules:
>
> - Return locations that the commenter asks be kept together during redistricting.
> - Return locations that the commenter asks be grouped with another location during redistricting.
> - Return locations that the commenter asks be separated from another location during redistricting.
> - Do not return locations outside the commenter's state.
> - Do not return any proposed legislative districts discussed by the commenter, only existing districts as of 2020.
> - Do not return roads or interstates mentioned by the commenter.

Since proposed legislative districts have no history of representing a community, we asked the Mistral model to only include legislative districts that were in place prior to the 2021-2022 redistricting cycle. These existing districts could reasonably have corresponded to a territorial community at the time that public comments were collected.

The prompts for individual location schema components are shown below in order of appearance, including the prompts for a location's name, administrative level, descriptive subareas, and relevance to a redistricting request:

> Best estimate of the type or administrative level of the location. Must be exclusive and decisive.

If the location's administrative level is unclear, return 'NA'.

Official and vernacular administrative levels for mentioned locations vary from state to state. Therefore, the factor levels available for administrative level prompt outputs were selected from the following set based on each state's relevant properties, including the presence of congressional districts and town/township/borough/city naming conventions for municipalities: "Landmark", "School", "Neighborhood", "Township", "Borough", "Town", "City", "School District", "County", "State House District", "State Senate District", "Congressional District", "Region", and "NA".

After finding the location's administrative level, return the canonical, administrative name of the location. Unless the location is a region, do not include any subareas in the name (i.e. return 'Washington County' if the comment mentions 'Northern Washington County' or 'Rural Washington County'). Location names should thus read like the following rules:

Naming rules:

- Landmark: use the landmark's canonical name.
- Neighborhood: use the neighborhood's canonical name.
- Township: use the municipality's canonical name (do not include 'township' in the name).
- Borough: use the municipality's canonical name.
- Town: use the municipality's canonical name.
- City: use the municipality's canonical name.
- School District: use '[Name] School District'.
- County: use '[Name] County'.
- State House District: use 'State House District [Number]'.
- State Senate District: use 'State Senate District [Number]'.
- Congressional District: use 'Congressional District [Number]'.
- Region: Name must exactly match one of the permitted region names.

Permitted region names: {{stateRegions}}

The naming rules for the prompt above were designed to constrain outputs for each administrative level to the naming structure of Census-defined places available through the `tigris` package in R (Walker 2025). The `stateRegions` variable was defined based on statewide cardinal direction subareas (i.e. "Northern Colorado") as well as common vernacular regions in each state (i.e. "Front Range", "Happy Valley", or "Atlanta Metro Area"). Although these sets of regions are not exhaustive, our inclusion of this variable as a prompt rule rather than a separate structured prompt allows the Mistral model to deviate from the provided set and include alternative region names when no clear choice is available among `stateRegions`.

> A subarea specific to the location, if explicitly mentioned by the commenter. If the commenter refers to the entirety of the location (i.e. 'Washington County'), return 'NA'. Proper names of a location or region (i.e. 'Westridge', 'South Bend', 'Northern California') do not imply a separate subarea mention (i.e. 'Western', 'Southern', and 'Northern'). Return 'NA' in these cases.

Subarea options included the following: "Northern", "Northeastern", "Eastern", "Southeastern", "Southern", "Southwestern", "Western", "Northwestern", "Central", "Unincorporated", "Rural", "Suburban", "Urban", "Downtown", "NA".

> The relevance of the location to a specific request by the commenter regarding a community of interest and a possible legislative district boundary choice. Locations should be regarded as 'contextual' when they provide background information or anecdotes that are tangential to genuine geographic configurations and the broader thesis of the comment. Relevant locations are those that a commenter requests be kept whole, grouped with other locations, or separated from other locations. The following are examples of when 'contextual' should be returned for a mentioned location.
>
> - I was at the meeting in Springfield last night. Here are my observations.
> - We need our community better represented in [State Capital City] and Washington DC.
> - Our neighborhood has been split, as have many neighborhoods across Washington County.
> - I came back to this community after attending college at [State University].

- My comments address several concerns I have over the districts in the western part of the state.

This relevance prompt and its associated rules were added based on edge cases observed during the process of reading randomly sampled comments. Since "contextual" locations are not part of a genuine geographic configuration requested by the commenter, they should not be included in downstream analysis steps, especially including the evaluation of district plan congruence with constituent opinion.

5.2 Stage Three - Location Relationship Extraction

For any comments identified as having multiple locations mentioned, a third set of `ellmer` prompts was used to extract the relationships between locations according to the commenter's request. Individual relationships were embedded in an array type `ellmer` output. Since the nature of these relationships are dependent on a commenter's potentially complex and multifaceted arguments and their approval or disapproval of either existing or proposed district maps, the prompt for this classifier included several examples and stipulations, as shown below:

> This commenter is providing public input about a legislative redistricting cycle, requesting that some geographic locations they mention either be kept together or separated into various districts based on their communities of interest. Return any pairwise location relationships requested by the commenter. Interpret the commenter's request, complaint, or concern. Do not treat the composition of an existing or proposed district as a desired relationship unless the commenter explicitly endorses or rejects that composition.
>
> The following are examples of a 'grouped' pairwise relationship between 'Springfield' and 'Washington County':
>
> - 'Vote no on the proposed map, which separates Springfield from Washington County.'
> - 'Springfield and Washington County have been separated. This is a clear effort to Gerrymander our community.'

- ‘Springfield and Washington County share similar economic interests.’
- ‘Why is Springfield in a different district than Washington County?’
- ‘Separating Springfield from Washington County is ridiculous.’
- ‘Please Keep Springfield together with Washington County.’
- ‘Don’t split Springfield from Washington County.’
- ‘Springfield should be reunited with Washington County.’

The following are examples of a ‘separated’ pairwise relationship between ‘Springfield’ and ‘Washington County’:

- ‘Vote no on the proposed map, which combines Springfield with Washington County.’
- ‘Springfield and Washington County have been combined. This is a clear effort to Gerrymander our community.’
- ‘Springfield and Washington County do not share similar economic interests.’
- ‘Why is Springfield in the same district as Washington County?’
- ‘Combining Springfield and Washington County is ridiculous.’
- ‘Please Keep Springfield separated from Washington County.’
- ‘Don’t group Springfield with Washington County.’
- ‘Springfield should be kept apart from Washington County.’

The prompt schema for each relationship included a first location name response, a second location name response, a relationship type response, and a district type response. Location name responses, while not tied to individual prompt descriptions, were constrained to an enum output, with factor levels derived from the composite information for each location extracted in the previous step of the pipeline. The straightforward relationship type prompting was as follows:

> Return the nature of the relationship between the two locations

Relationship types were constrained to be either “grouped”, “separated”, or “unclear”. District type prompting consisted of the following:

> Return the legislative district type relevant to this pairwise relationship request, if mentioned by the commenter. If the relevant district types are unclear or if multiple types are relevant, return 'NA'.

District types relevant to the location relationship were constrained to be one or more of the following: "State House", "State Senate", "Congressional", or "NA" when no district type was listed or when multiple types were being discussed simultaneously.

5.3 Validation

For each state, 50 comments screened for the mention of communities of interest were randomly sampled and annotated by hand. These annotated comments followed the same rules as the prompting architecture described above, and from location mentions and requests among pairs of locations, a "Ground Truth" set of comments was constructed to validate `ellmer` output against. This ground truth set includes 592 mentioned locations and 329 groups of locations. For each validation test, two comparison sets of locations or requests were compared to a ground truth set using linear regression with robust standard errors under the following model:

$$Y = \beta_0 + State$$

Where Y is a precision or recall contrast between the two methods against the ground truth at the unit level, β_0 is an intercept term, and $State$ is a state-wise fixed effect. In the context of our analysis, precision is the number of correctly identified units as a proportion of all identified units, while recall is the number of correctly identified units as a proportion of ground truth units. Regression coefficient tests below are against the null hypothesis that the intercept term is indistinguishable from 0, except in the cases of location level precision because ground truth and comparison unit sets are not necessarily of the same size, prohibiting the use of unit-wise contrast. In these cases, regression coefficient tests compare against a null hypothesis that a binary coefficient differentiating the two comparison methods is indistinguishable from 0.

5.3.0.1 Location Recognition Validation Table 1 explores the efficacy of the relevance field prompting. Removing contextual locations offered statistically significant increases to precision while only slightly diminishing recall, so this field was deemed useful for the purposes of consolidating downstream location request options.

Table 1: Unit contrast regression tests of full baseline location set (Method A) compared to a baseline location with contextual locations removed (Method B)

Test	Unit	Method A	Method B	Coefficient	P-Value
Location Recall	Location	0.863	0.834	-0.005	0.318
Location Recall	Comment	0.844	0.835	-0.007	0.319
Location Precision	Location	0.689	0.752	0.063	0.008
Location Precision	Comment	0.681	0.740	0.059	0.099

We also investigated precision and recall across multiple iterations of each location recognition task. Table 2 compares a method filtering mentioned locations that were mentioned at least twice across three iterations to locations mentioned in a single iteration. Table 3 compares a set for which mentions were mentioned in all three iterations to the same single iteration. Restricting locations to majority or unanimous mentions was associated with some small tradeoffs between precision and recall, with no definitive advantage to the task's broader accuracy, so we concluded that single-iteration extraction tasks were sufficient for downstream analysis.

Table 2: Unit contrast regression tests of single iteration locations (Method A) compared to locations mentioned across a majority of three iterations (Method B)

Test	Unit	Method A	Method B	Coefficient	P-Value
Location Recall	Location	0.834	0.841	0.014	0.257
Location Recall	Comment	0.835	0.839	0.013	0.128
Location Precision	Location	0.752	0.756	0.004	0.869
Location Precision	Comment	0.740	0.753	0.013	0.706

Table 3: Unit contrast regression tests of single iteration locations (Method A) compared to locations mentioned across unanimously three iterations (Method B)

Test	Unit	Method A	Method B	Coefficient	P-Value
Location Recall	Location	0.834	0.814	-0.023	0.058
Location Recall	Comment	0.835	0.826	-0.009	0.389
Location Precision	Location	0.752	0.797	0.045	0.055
Location Precision	Comment	0.740	0.797	0.057	0.108

The baseline prompting architecture, as described above, was developed iteratively, succeeding several other variants as the idiosyncrasies of the `ellmer` package and its interactions with the Mistral model were discovered. Table 4 below shows a comparison between the baseline prompting architecture for location recognition above and a simpler prompting architecture lacking rules at the system prompt and contextual location identification levels. This simpler architecture was not statistically distinguishable from the more verbose architecture, but the intercept term differences for the latter were signed in the direction of higher precision and recall, leading us to designate it as our baseline.

Table 4: Unit contrast regression tests of baseline prompting architecture (Method A) compared to a simpler prompting architecture with fewer prompt examples (Method B)

Test	Unit	Method A	Method B	Coefficient	P-Value
Location Recall	Location	0.846	0.826	-0.042	0.060
Location Recall	Comment	0.847	0.816	-0.056	0.037
Location Precision	Location	0.760	0.775	0.015	0.535
Location Precision	Comment	0.761	0.768	0.007	0.848

Lastly, Table 5 compares a simpler prompting architecture with and without a system prompt. Although not statistically significant, substantively large increases to precision under the inclusion of a system prompt drove us to retain the prompt in our baseline architecture.

Table 5: Unit contrast regression tests of simpler prompting architecture without a system prompt (Method A) compared to a similar prompting architecture with a system prompt (Method B)

Test	Unit	Method A	Method B	Coefficient	P-Value
Location Recall	Location	0.878	0.853	0.014	0.468
Location Recall	Comment	0.843	0.840	0.006	0.838
Location Precision	Location	0.680	0.722	0.045	0.059
Location Precision	Comment	0.679	0.713	0.034	0.340

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